

Customer Shopping Behavior Intelligence

End-to-End Data Analytics Project | Python • SQL • Power BI

1. Project Context

In highly competitive retail environments, understanding what drives customer spending, loyalty, and product performance is critical.

This project analyzes 3,900 customer transactions to uncover behavioral insights and generate data-driven commercial strategies.

2. Business Objectives

- Identify high revenue customer segments
- Evaluate discount effectiveness
- Analyze subscription value contribution
- Understand demographic spending patterns
- Detect product pricing dependency

3. Dataset Overview

- Transactions: 3,900
- Features: 18
- Categories: Accessories, Clothing, Footwear, Outerwear

4. Data Preparation & Feature Engineering (Python)

- Data cleaning and preprocessing were performed using Pandas to ensure data reliability .
- Imputed missing review ratings using the median rating per product category.
- Renamed columns into snake case format for consistency.
- Created age group variable through age binning.
- Derived purchase frequency days metric.
- Removed redundant column promocode used.
- Cleaned dataset was loaded into PostgreSQL for structured querying.

```
[23]: # Loading the dataset using pandas
```

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
[24]: df.head()
```

```
[24]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	V
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	I
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	I

```
[ ]:
```

```
[28]: # Imputing missing values in Review Rating column with the median rating of the product category
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
[29]: df.isnull().sum()
```

```
[29]: Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD) 0
Location           0
Size              0
Color             0
Season            0
Review Rating      0
Subscription Status 0
Shipping Type      0
Discount Applied   0
Promo Code Used    0
Previous Purchases 0
Payment Method     0
Frequency of Purchases 0
dtype: int64
```

```
[30]: # Renaming columns according to snake casing for better readability and documentation
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
[31]: df.columns
```

```
[31]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
[34]: # create new column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

[35]: df[['discount_applied', 'promo_code_used']].head(10)
```

```
[35]:
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
[36]: (df['discount_applied'] == df['promo_code_used']).all()
```

```
[36]: np.True_
```

```
[37]: # Dropping promo code used column

df = df.drop('promo_code_used', axis=1)
```

5. Data Preparation & Feature Engineering

- Median imputation for missing ratings
- Created age cohorts and purchase frequency metrics
- Customer lifecycle segmentation (New / Returning / Loyal)
- MySQL integration for scalable querying

6. Key Insights

- Young Adults are the highest revenue contributors
- Express shipping users show higher purchase intent
- Certain products show ~50% discount dependency
- Loyal customers dominate the customer base
- Non-subscribers generate higher total revenue due to population size

7. Business Analysis Using SQL

- **Revenue Contribution by Gender** - Male customers generated significantly higher total revenue compared to female customers

Result Grid		
	gender	revenue
▶	Male	157890
	Female	75191

High-Spending Discount Users

- **High-Spending Discount Users** - 839 customers were identified who used discounts but still spent above average.

Result Grid		
	customer_id	purchase_amount
	1652	80
	1654	93
	1656	81
	1659	66
	1662	86
	1667	64
	1671	73
	1673	73
	1674	62
	1676	90

- **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	avg review rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

- **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

Result Grid				
	subscription_status	total_customers	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

- **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

- **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

Result Grid			Filter Rows:	
	shipping_type	ROUND(AVG(purchase_amount),2)		
▶	Express	60.48		
	Standard	58.46		

- **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	number_of_customers
▶	Loyal	3116
	Returning	701
	New	83

- **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	149
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

- **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
►	Yes	958
	No	2518

- **Revenue by Age Group** – Calculated total revenue contribution of each age group.

Result Grid		Filter Rows:
	age_group	total_revenue
►	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

8. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



9. Strategic Recommendations

- Improve subscription benefits to increase CLV
- Optimize discount allocation using product elasticity signals
- Target high-value demographics for marketing ROI uplift
- Promote top-rated products for conversion improvement

- Strengthen loyalty incentives for retention growth

10. Project Takeaways

This project demonstrates strong capability in data cleaning, SQL analytics, dashboard storytelling, KPI interpretation, and business strategy translation.